HEALTHIER, LONGER, BETTER LIVES

Medical Receipt OCR Benchmark

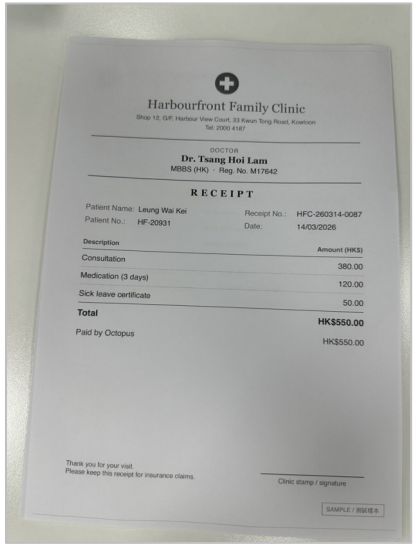
Can Azure OpenAI replace hand-written extraction rules?

9 receipts · 6 fields · 4 approaches

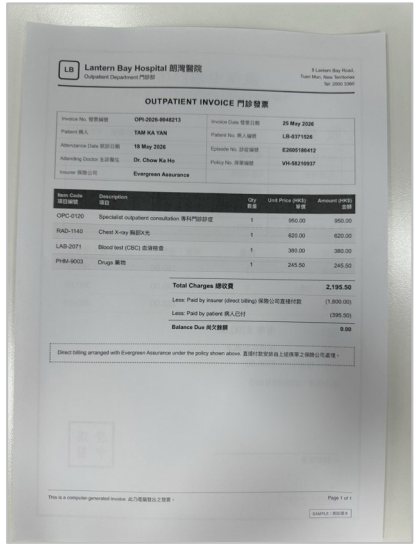
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What We Tested

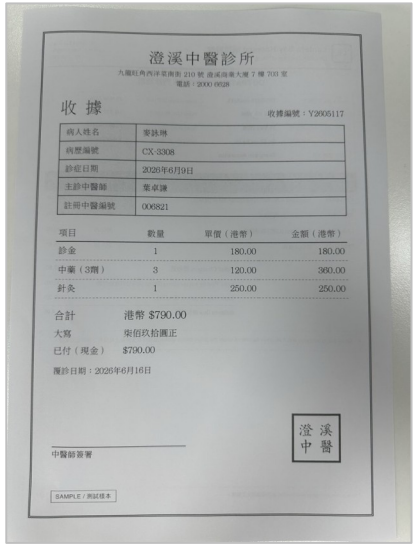
9 receipts with different layouts and languages. Each approach reads the text, then finds 6 fields.



English



Bilingual



Chinese

Approach	1. Read text	2. Find fields
PaddleOCR Light	Open-source OCR	Rules
PaddleOCR Heavy	Open-source OCR	Rules
Azure DI	Azure OCR	Rules
Azure DI + AOAI	Azure OCR	Azure OpenAI

Fields: doctor, registration no., patient, receipt no., service date, total.
51 fields scored.

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Results: Same Accuracy, Different Effort

Rules reached 88.2% only after manual tuning. DI + AOAI reached 88.2% with its prompt unchanged.

Approach	Original rules	Improved rules*	Time per receipt
PaddleOCR Light	37.3% (19/51)	**88.2%** (45/51)	13.9 s
PaddleOCR Heavy	29.4% (15/51)	64.7% (33/51)	53.3 s
Azure DI	41.2% (21/51)	**86.3%** (44/51)	3.8 s
Azure DI + AOAI	**88.2%** (45/51)	No rules (prompt unchanged)	6.2 s

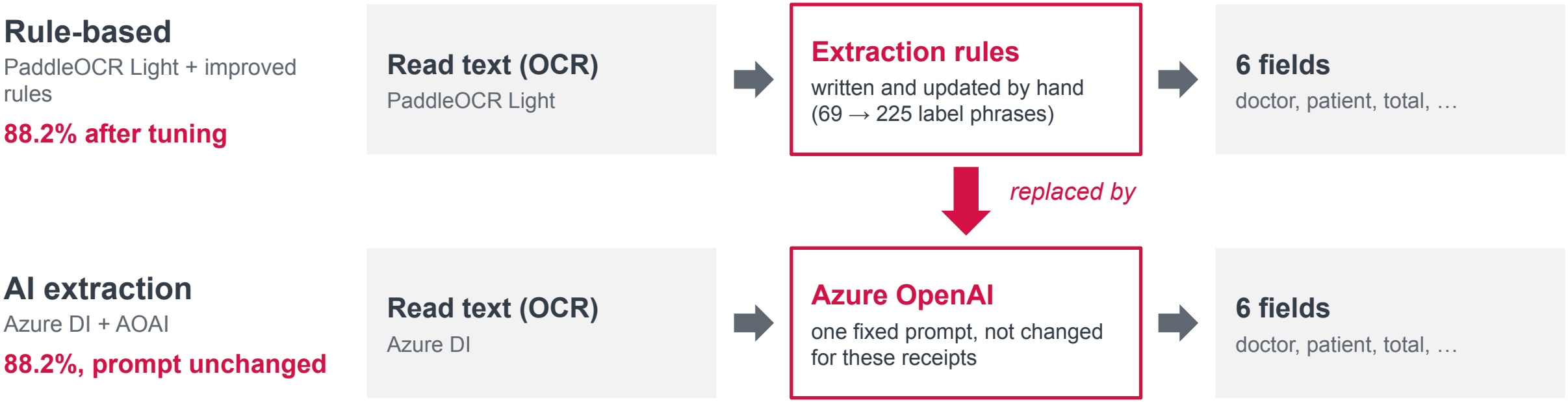
** Tuned using these 9 receipts, so not a measure of accuracy on new receipts. Accuracy = correct fields out of 51. All approaches processed all 9 receipts.*

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AOAI Replaces the Manual Rule Layer

Both pipelines read the text the same way. Only the step that finds the fields changes.



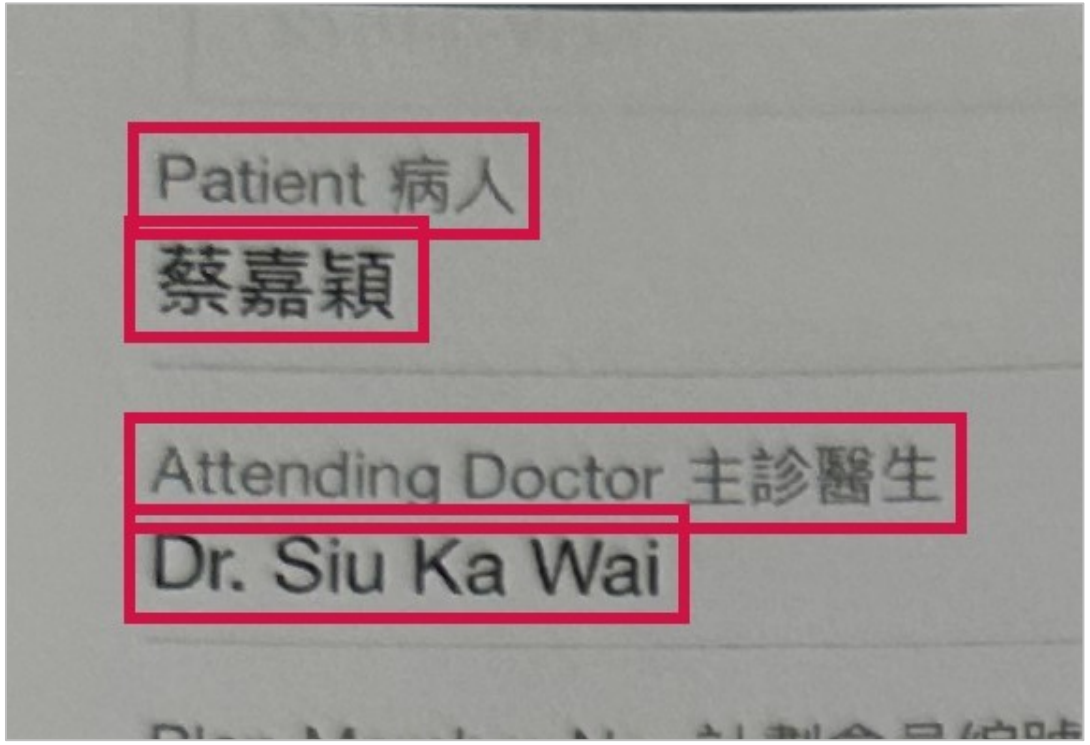
AOAI replaces the manual rule layer. It still makes mistakes (6 of 51 fields wrong) and has not yet been tested on a large set of unseen receipts.

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Example 1: The OCR Was Right, the Rules Were Wrong

Both OCR engines read the names. The original rules took the label 病人 as the patient's name.



Field	Correct	Original rules	Improved rules	DI + AOAI
Patient	蔡嘉穎	✗ 病人	✓ 蔡嘉穎	✓ 蔡嘉穎
Doctor	Dr. Siu Ka Wai	✗ not found	✓ Dr. Siu Ka Wai	✓ Dr. Siu Ka Wai

- Original rules: took the Chinese label 病人 ("patient") as the patient's name.
- Improved rules: fixed by adding bilingual labels by hand.
- AOAI: correct with the same prompt; no change needed.

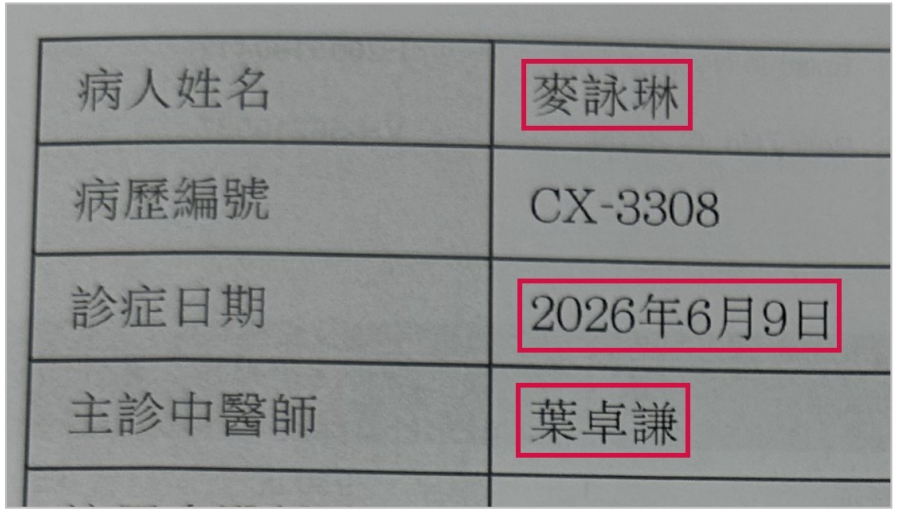
Bilingual medical centre receipt. Rules shown on PaddleOCR Light text; Azure DI with the original rules made the same patient error.

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Example 2: Reaching 88.2% With Rules Took Manual Work

Each new language or label wording had to be added by hand, and one change broke another receipt.



Chinese medicine receipt: original rules 0/6 fields, improved rules 6/6, DI + AOAI 6/6 (prompt unchanged).

69 → 225

label phrases in the rules, added by hand after reviewing the errors

369303 → 380048

vet invoice receipt number: correct before the changes, wrong after (a regression)

New layouts

will likely need more rule changes and a retest of every receipt

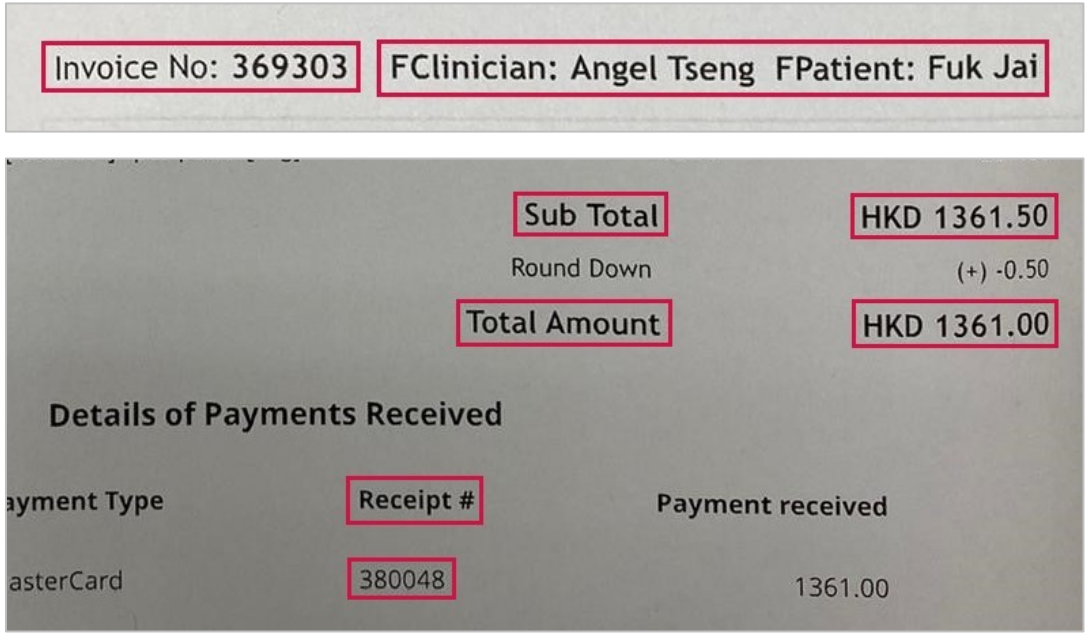
DI + AOAI needed none of these changes for these receipts. Whether that holds for new layouts still needs to be tested.

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Example 3: AOAI Still Makes Mistakes

On the vet invoice, the improved rules and AOAI each got a different field wrong.



Field	Correct	Original rules	Improved rules	DI + AOAI
Receipt no.	369303	✓ 369303	✗ 380048	✓ 369303
Patient	Fuk Jai	✗ not found	✗ not found	✓ Fuk Jai
Total	1361	✓ 1361.0	✓ 1361.0	✗ 1361.5

- Improved rules: picked the payment receipt number.
- Rules: the OCR read "FPatient:", so the patient label was missed.
- AOAI: picked the sub total instead of the total.

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What It Takes to Reach 88.2%

Same accuracy. The difference is the manual work to get there, and to keep it there.

	PaddleOCR Light + improved rules	Azure DI + AOAI
Accuracy (9 receipts)	**88.2% (45/51)**	**88.2% (45/51)**
Effort to get there	Rules rewritten by hand after reviewing errors on these receipts	None: prompt unchanged
Extraction logic	69 → 225 label phrases to maintain	One fixed prompt
New receipt layouts	Likely more rule updates and a regression test of every receipt	Possibly no change; not yet tested
Errors seen	6 wrong fields, incl. one regression caused by tuning	6 wrong fields, e.g. the vet invoice total
Time and running cost	13.9 s, own compute, no API fee	6.2 s, Azure fees per receipt (to confirm)

The main question is not API price but who maintains the extraction logic. Lower maintenance for AOAI is implied by this benchmark, not yet proven.

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Pure OCR + Rules vs. Azure DI + AOAI: Pros and Cons

Both reached 88.2%. Points marked "Measured" come from this benchmark; the rest are expected, not yet tested.

Pure OCR + Rules (PaddleOCR Light)

Pros

- Full control over the extraction logic.
- No API fees: open-source OCR on our own compute.
- Predictable on receipt formats the rules already know.
- **Measured:** 88.2% after careful tuning (from 37.3%).

Cons

- **Measured:** rules expanded by hand (69 → 225 label phrases).
- New languages, labels and layouts need more rule changes.
- **Measured:** a rule update broke a receipt that was already correct.
- Accuracy on unseen receipt formats is unknown.

Azure DI + AOAI

Pros

- **Measured:** 88.2% with the prompt unchanged.
- Reduces the need to write detailed extraction rules.
- **Measured:** 6/6 fields on both the bilingual and Chinese examples.
- **Measured:** faster, 6.2 s vs 13.9 s per receipt.

Cons

- Needs Azure services and per-request API fees.
- **Measured:** wrong value even though the OCR text was correct.
- Results may vary, so validation and error handling are needed.
- Accuracy and maintenance effort on unseen receipts not yet proven.

The decision is not only accuracy or API cost: it is how much development and ongoing maintenance we want to own.

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Conclusion and Next Steps

The real trade-off: maintain extraction rules ourselves, or let AI replace that layer.

- PaddleOCR Light + improved rules and Azure DI + AOAI both reached 88.2% on this benchmark.
- PaddleOCR Light needed manual rule tuning to get there. Azure DI + AOAI reached the same accuracy without changing its prompt.
- **So the choice is whether to keep writing and maintaining extraction rules, or to use AI for that layer.**

1. Freeze both

Lock the improved rules and the AOAI prompt.

2. Test on unseen receipts

30+ real, anonymised receipts. Measure accuracy and count the rule changes needed.

3. Then compare cost

Compute, Document Intelligence pages and gpt-5-mini tokens.

Limitation: 9 receipts, mostly printed mock receipts; the improved rules were tuned on them. AOAI still makes mistakes and has not been tested on a large unseen set.

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